

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29-06-2025
Team ID	23A51A05H9
Project Name	Leave Approval System
Maximum Marks	4 Marks

Technical Architecture:

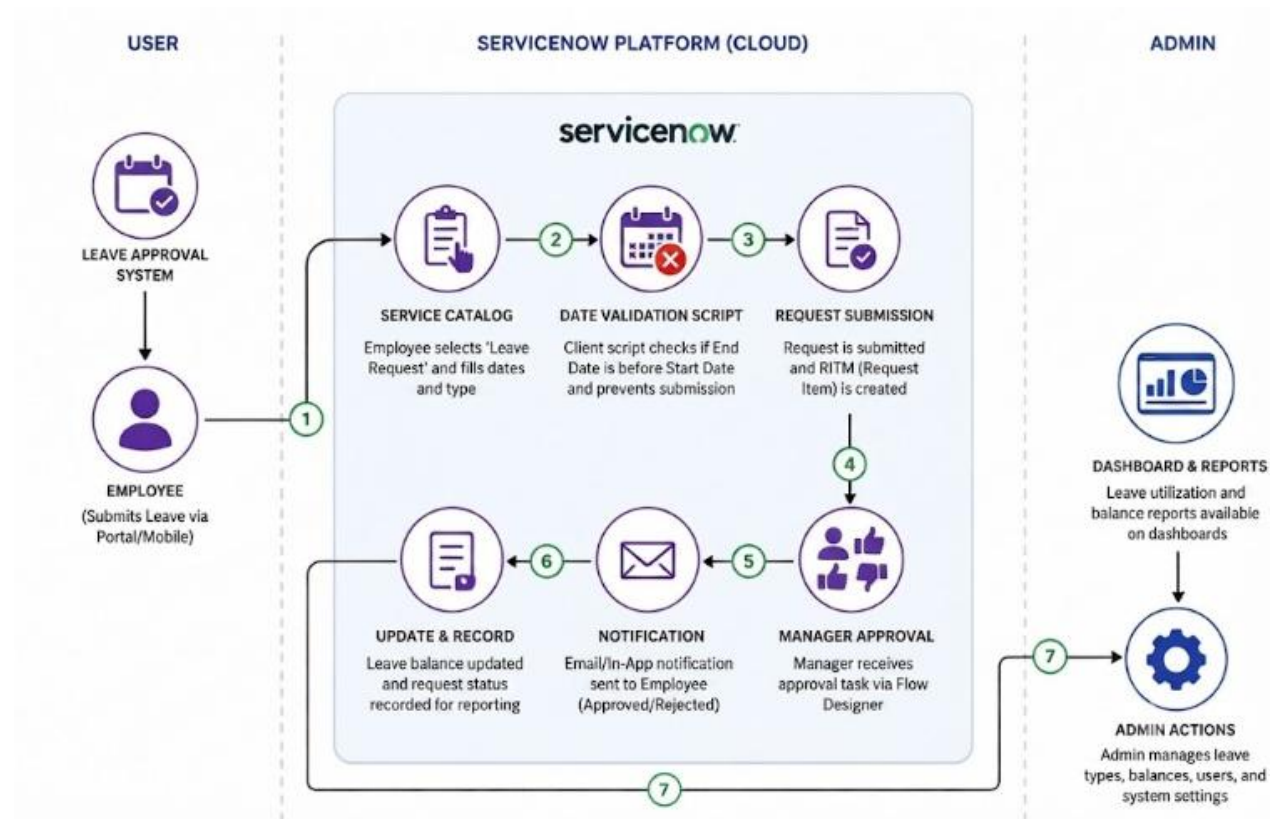


Table-1 : Components & Technologies:

<u>S.No</u>	Component	Description	Technology
1	User Interface	Interface for employees to submit leave requests and choose dates/types	ServiceNow Service Portal / Employee Center, Form UI
2	Application Logic-1	Instant form validation ensuring the End Date is not earlier than the Start Date	Catalog Client Scripts / UI Policies (JavaScript)
3	Application Logic-2	Server-side user evaluation (auto-populating the employee's direct manager on the form)	Script Includes, GlideAjax
4	Application Logic-3	Automated manager multi-tier approval routing and status updates	Flow Designer / Workflow Studio
5	Database	Stores leave application data, request choices, and approval states	Custom Scoped ServiceNow Table (x_snc_leave_app_leave_request)
6	Cloud Database	Secure, cloud-based transactions and logs managed by ServiceNow	ServiceNow Cloud Database (MariaDB/MySQL Instance)
7	File Storage	Stores supplementary medical certificates or leave justification documents	ServiceNow Attachments (sys_attachment)
8	Notifications	Triggers automated email updates to employees when leave is approved or rejected	ServiceNow Notification Engine / Flow Designer Email Action
9	In-App Alerts	Sends immediate operational alerts and red-dot badge updates directly inside the system banner	ServiceNow Mobile/Push Notifications & Live Feed Alerts
10	Reporting & Analytics	Provides HR and management with visibility into leave trends, team utilization, and balance history	ServiceNow Reports & Dashboards

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Client-side scripting and customization used.	JavaScript, GlideAjax
2.	Security Implementations	Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission	ServiceNow ACLs, User Roles, HTTPS, IAM Controls
3.	Scalable Architecture	ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes	ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform
4.	Availability	High availability through ServiceNow cloud infrastructure managed services	ServiceNow Cloud Infrastructure with continuous monitoring and Redundant Data Centers
5.	Performance	Fast ticket booking process through server-side processing, cloud-hosted services	GlideAjax, Script Includes, ServiceNow Database Optimization

References:

<https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html>

<https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html>

https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html

<https://developer.servicenow.com>

<https://www.servicenow.com/community/>